

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihal Produk: **Bis(2-chloroethyl) ether**  
 Product Description: **Bis(2-chloroethyl) ether**  
 Cat No. : A14234  
 Synonyms Bis-2-chloroethyl ether  
 CAS No 111-44-4  
 Molecular Formula C<sub>4</sub> H<sub>8</sub> Cl<sub>2</sub> O

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

**Company**

Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**Supplier**

E-mail address Enquiry.my@thermofisher.com

**Emergency Telephone Number**

Tel: +03-5525 7888  
 CHEMTREC Malaysia **1-800-815-308** (Malay)  
 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                    |                   |
|------------------------------------|-------------------|
| Flammable liquids                  | Category 3 (H226) |
| Acute oral toxicity                | Category 2 (H300) |
| Acute dermal toxicity              | Category 1 (H310) |
| Acute Inhalation Toxicity - Vapors | Category 2 (H330) |
| Carcinogenicity                    | Category 2 (H351) |

**Label Elements**


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## Signal Word

## Danger

### Hazard Statements

H226 - Flammable liquid and vapor

H300 + H310 + H330 - Fatal if swallowed, in contact with skin or if inhaled

H351 - Suspected of causing cancer

### Precautionary Statements

#### Prevention

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P240 - Ground and bond container and receiving equipment

P241 - Use explosion-proof electrical/ ventilating/ lighting equipment

P242 - Use non-sparking tools

P243 - Take action to prevent static discharges

P262 - Do not get in eyes, on skin, or on clothing

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P284 - Wear respiratory protection

#### Response

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P310 - Immediately call a POISON CENTER or doctor

P330 - Rinse mouth

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

P362 + P364 - Take off contaminated clothing and wash it before reuse

#### Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

P405 - Store locked up

#### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

### Other Hazards

Lachrymator (substance which increases the flow of tears)

Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                | CAS No   | Weight % |
|--------------------------|----------|----------|
| Bis(2-chloroethyl) ether | 111-44-4 | > 99     |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### Eye Contact

Immediate medical attention is required. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.

#### Skin Contact

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Immediate medical attention is required.

#### Ingestion

Call a physician immediately. Clean mouth with water.

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|                                           |                                                                                                                                                  |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                         | Remove from exposure, lie down. Remove to fresh air. If not breathing, give artificial respiration. Immediate medical attention is required.     |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

**Most important symptoms and effects, both acute and delayed**

Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

**Indication of any immediate medical attention and special treatment needed**

**Notes to Physician** Treat symptomatically. Symptoms may be delayed.

## SECTION 5: FIREFIGHTING MEASURES

**Extinguishing media**

**Suitable Extinguishing Media**

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam. Water mist may be used to cool closed containers.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Special hazards arising from the substance or mixture**

Flammable. Combustible material. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

**Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Hydrogen chloride gas.

**Advice for fire-fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

**Personal Precautions, Protective Equipment and Emergency Procedures**

Remove all sources of ignition. Take precautionary measures against static discharges.

**Environmental precautions**

See Section 12 for additional Ecological Information.

**Methods and Material for Containment and Cleaning Up**

Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Wear self-contained breathing apparatus and protective suit. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. Do not let this chemical enter the environment.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

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## Precautions for Safe Handling

Do not breathe mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Do not ingest. If swallowed then seek immediate medical assistance. Handle product only in closed system or provide appropriate exhaust ventilation. Use spark-proof tools and explosion-proof equipment. Use only non-sparking tools. Keep away from open flames, hot surfaces and sources of ignition. Take precautionary measures against static discharges.

## Conditions for Safe Storage, Including any Incompatibilities

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep away from heat, sparks and flame. Flammables area. Keep container tightly closed in a dry and well-ventilated place.

## Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component                | Malaysia | ACGIH TLV                          | OSHA PEL                                                                                                                                                                                  |
|--------------------------|----------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bis(2-chloroethyl) ether |          | TWA: 5 ppm<br>STEL: 10 ppm<br>Skin | (Vacated) TWA: 5 ppm<br>(Vacated) TWA: 30 mg/m <sup>3</sup><br>Ceiling: 15 ppm<br>Ceiling: 90 mg/m <sup>3</sup><br>(Vacated) STEL: 10 ppm<br>(Vacated) STEL: 60 mg/m <sup>3</sup><br>Skin |

| Component                | European Union | The United Kingdom | Germany                                                                                                                                                                                                                                                        |
|--------------------------|----------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bis(2-chloroethyl) ether |                |                    | TWA: 0.5 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 3 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 0.5 ppm (8 Stunden). MAK<br>TWA: 3 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 1 ppm<br>Höhepunkt: 6 mg/m <sup>3</sup><br>Haut |

### Exposure Controls

#### Engineering Measures

Use explosion-proof electrical/ventilating/lighting equipment. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

#### Personal protective equipment

##### Eye Protection

Goggles

##### Hand Protection

Protective gloves

##### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

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Remove gloves with care avoiding skin contamination.

## Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

## Recommended Filter type:

Organic gases and vapours filter Type A Brown conforming to EN14387

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

#### Appearance

Colorless

#### Physical State

Liquid

#### Odor

pungent

#### Odor Threshold

No data available

#### pH

No information available

#### Melting Point/Range

-52 °C / -61.6 °F

#### Softening Point

No data available

#### Boiling Point/Range

178.5 °C / 353.3 °F

@ 760 mmHg

#### Flash Point

55 °C / 131 °F

Method - No information available

#### Evaporation Rate

No data available

#### Flammability (solid,gas)

Not applicable

Liquid

#### Explosion Limits

Lower 0.8

#### Vapor Pressure

1.1 mbar @ 20 °C

#### Vapor Density

4.93 (Air = 1.0)

(Air = 1.0)

#### Specific Gravity / Density

1.220

#### Bulk Density

Not applicable

Liquid

#### Water Solubility

Insoluble

#### Solubility in other solvents

No information available

#### Partition Coefficient (n-octanol/water)

##### Component

log Pow

Bis(2-chloroethyl) ether

1.12

#### Autoignition Temperature

369 °C / 696.2 °F

#### Decomposition Temperature

No data available

#### Viscosity

No data available

#### Explosive Properties

explosive air/vapour mixtures possible

#### Oxidizing Properties

No information available

#### Molecular Formula

C4 H8 Cl2 O

#### Molecular Weight

143.02

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## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under normal conditions.

### Possibility of Hazardous Reactions

**Hazardous Polymerization** No information available.  
**Hazardous Reactions** No information available.

### Conditions to Avoid

Keep away from open flames, hot surfaces and sources of ignition. Incompatible products.

### Incompatible Materials

None known.

### Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen chloride gas.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

(a) acute toxicity;  
Oral Category 2  
Dermal Category 1  
Inhalation Category 2

| Component                | LD50 Oral               | LD50 Dermal                 | LC50 Inhalation               |
|--------------------------|-------------------------|-----------------------------|-------------------------------|
| Bis(2-chloroethyl) ether | LD50 = 75 mg/kg ( Rat ) | LD50 = 870 mg/kg ( Rabbit ) | LC50 = 1.464 mg/L ( Rat ) 4 h |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;  
Respiratory No data available  
Skin No data available

(e) germ cell mutagenicity; No data available

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(f) carcinogenicity; Category 2  
There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; No data available

Other Adverse Effects The toxicological properties have not been fully investigated.

Symptoms / effects, both acute and delayed Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity effects Do not empty into drains. .

| Component                | Freshwater Fish                                    | Water Flea                                       | Freshwater Algae | Microtox |
|--------------------------|----------------------------------------------------|--------------------------------------------------|------------------|----------|
| Bis(2-chloroethyl) ether | LC50: = 600 mg/L, 96h static (Lepomis macrochirus) | LC50: 170 - 330 mg/L, 48h Static (Daphnia magna) |                  |          |

Persistence and degradability  
Persistence Persistence is unlikely.

Bioaccumulative potential Bioaccumulation is unlikely

| Component                | log Pow | Bioconcentration factor (BCF) |
|--------------------------|---------|-------------------------------|
| Bis(2-chloroethyl) ether | 1.12    | 11 L/kg                       |

Mobility in soil Spillage unlikely to penetrate soil. The product is insoluble and sinks in water. . Is not likely mobile in the environment due its low water solubility.

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

Other adverse effects No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

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## Waste treatment methods

### Waste from Residues/Unused Products

Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous Keep product and empty container away from heat and sources of ignition

### Other Information

Waste codes should be assigned by the user based on the application for which the product was used Do not flush to sewer Can be landfilled or incinerated, when in compliance with local regulations

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

UN-No UN1916  
Hazard Class 6.1  
Subsidiary Hazard Class 3  
Packing Group II  
Proper Shipping Name 2,2`-DICHLORODIETHYL ETHER

### Road and Rail Transport

UN-No UN1916  
Hazard Class 6.1  
Subsidiary Hazard Class 3  
Packing Group II  
Proper Shipping Name 2,2`-DICHLORODIETHYL ETHER

### IATA

UN-No UN1916  
Hazard Class 6.1  
Subsidiary Hazard Class 3  
Packing Group II  
Proper Shipping Name 2,2`-DICHLORODIETHYL ETHER

Special Precautions for User No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories X = listed

| Component                | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|--------------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Bis(2-chloroethyl) ether | 203-870-1 | X    | -   | X     | X    | X    | X     | X    | KE-10105 |

### National Regulations

#### Persistent Organic Pollutant Ozone Depletion Potential

This product does not contain any known or suspected substance

This product does not contain any known or suspected substance

| Component                | Persistent Organic Pollutant | Ozone Depletion Potential | Pesticides Act 1974 |
|--------------------------|------------------------------|---------------------------|---------------------|
| Bis(2-chloroethyl) ether |                              |                           | X                   |



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## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Prepared By**

**Revision Date**

**Revision Summary**

Health, Safety and Environmental Department

25-Mar-2025

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

### **Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**